RAJALAKSHMI ENGINEERING COLLEGE

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C2

Started on	Thursday, 17 September 2026, 8:24 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:25 AM
Time taken	38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

#define MAX_N 100005

int sequence[MAX_N];
int tails[MAX_N];

// Binary search to find the first element strictly greater than key (upper_bound)
int upper_bound(int arr[], int length, int key) {
    int low = 0, high = length;
    while (low < high) {
        int mid = low + (high - low) / 2;
        if (arr[mid] <= key) {
            low = mid + 1;
        } else {
            high = mid;
        }
    }
    return low;
}

int main() {
    int n;

    // Read the number of elements
    if (scanf("%d", &n) != 1) return 0;

    // Read the sequence values
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &sequence[i]) != 1) return 0;
    }

    int len = 0; // Tracks the length of the longest non-decreasing subsequence

    for (int i = 0; i < n; i++) {
        int idx = upper_bound(tails, len, sequence[i]);

        tails[idx] = sequence[i];

        if (idx == len) {
            len++; // Expanded the length of the subsequence
        }
    }

    // Output the maximum length found
    printf("%d\n", len);

    return 0;
}

```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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